As a data scientist, I sometimes want to explore different types of machine learning algorithms for different problems. For my reference, I created a list of the majority of ML algorithms. Hopefully this list can help others as well. The algorithms are grouped by category.

1. Supervised Learning

Supervised learning algorithms learn from labeled data, where the input-output pairs are known.

a. Regression Algorithms

- Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression
- Elastic Net Regression
- Support Vector Regression (SVR)

- Decision Tree Regression
- Random Forest Regression
- Gradient Boosting Regression (e.g., XGBoost, LightGBM, CatBoost)
- Bayesian Regression
- K-Nearest Neighbors (KNN) Regression

b. Classification Algorithms

- Logistic Regression
- Support Vector Machines (SVM)
- K-Nearest Neighbors (KNN) Classification
- Decision Trees
- Random Forest Classification
- Gradient Boosting Machines (GBM)
- AdaBoost
- XGBoost
- LightGBM
- CatBoost

- Naive Bayes
- Neural Networks (e.g., Multilayer Perceptron)
- Quadratic Discriminant Analysis (QDA)
- Linear Discriminant Analysis (LDA)

2. Unsupervised Learning

Unsupervised learning algorithms learn from unlabeled data, identifying patterns and structures.

a. Clustering Algorithms

- K-Means
- Hierarchical Clustering
- DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
- Mean Shift
- Gaussian Mixture Models (GMM)
- BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies)

- Affinity Propagation

b. Dimensionality Reduction Algorithms

- Principal Component Analysis (PCA)
- Independent Component Analysis (ICA)
- t-Distributed Stochastic Neighbor Embedding (t-SNE)
- Linear Discriminant Analysis (LDA)
- Factor Analysis
- Non-Negative Matrix Factorization (NMF)
- UMAP (Uniform Manifold Approximation and Projection)

3. Semi-Supervised Learning

Semi-supervised learning algorithms work with a small amount of labeled data and a large amount of unlabeled data.

- Self-training
- Co-training

- Generative Adversarial Networks (GANs) for Semi-Supervised Learning
- Graph-based Semi-Supervised Learning

4. Reinforcement Learning

Reinforcement learning algorithms learn by interacting with an environment to maximize cumulative rewards.

- Q-Learning
- SARSA (State-Action-Reward-State-Action)
- Deep Q-Networks (DQN)
- Policy Gradient Methods
- Actor-Critic Methods
- Deep Deterministic Policy Gradient (DDPG)
- Proximal Policy Optimization (PPO)
- Trust Region Policy Optimization (TRPO)

5. Ensemble Learning

Ensemble learning algorithms combine multiple models to improve performance.

- Bagging (Bootstrap Aggregating)
- Random Forest
- Boosting (e.g., AdaBoost, Gradient Boosting)
- Stacking
- Voting Classifier
- Blending

6. Neural Networks and Deep Learning

Deep learning algorithms involve multiple layers of neural networks.

- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- Long Short-Term Memory Networks (LSTM)
- Gated Recurrent Unit (GRU)
- Autoencoders

- Generative Adversarial Networks (GANs)
- Transformer Networks
- BERT (Bidirectional Encoder Representations from Transformers)
- GPT (Generative Pre-trained Transformer)

7. Probabilistic and Statistical Models

These algorithms are based on statistical methods and probabilistic reasoning.

- Bayesian Networks
- Hidden Markov Models (HMM)
- Markov Chain Monte Carlo (MCMC)
- Gaussian Processes

8. Instance-Based Learning

These algorithms memorize instances from the training data to make predictions.

- K-Nearest Neighbors (KNN)
- Locally Weighted Learning
- Case-Based Reasoning

9. Genetic Algorithms and Evolutionary Computing

These algorithms are inspired by the process of natural selection.

- Genetic Algorithms (GA)
- Genetic Programming (GP)
- Evolutionary Strategies (ES)
- Differential Evolution (DE)

10. Hybrid Methods

Combining multiple types of algorithms for better performance.

- Neural Evolution (combination of neural networks and genetic algorithms)

- Neuro-Fuzzy Systems (combination of neural networks and fuzzy logic)